

Supplementary Material for “The Wrong Unit of Uncertainty: Adaptive Design-Aware Conformal Inference for Repeated Cross-National Surveys”

The Authors

This online supplement collects secondary material referenced from the main manuscript. All content, numbers, and figures are reproduced unchanged; only figure numbers carry an “S” prefix.

S1 The finite- K mechanism and the development grid

Development: why the third gate is needed. On a Gaussian grid with known truth ($K \in \{30, 60, 120, 240\}$, ρ up to 1.8), the selector transitions from PCB to deconvolution at ρ_0 and to conservative at large ρ ; but a naive deconvolution that omits gate B *undercovers* at small K —coverage falls to 0.75 at $K = 30$ in the transition zone—because the target scale is estimated from few countries (Figure S3). Adding the finite- K -safe scale lifts this to 0.82, and the reliability gate B lifts the deployed worst cell to 0.862 while abstaining to conservative wherever the remainder cannot be bounded. These four transition cells at 0.862–0.878 are the development evidence that *motivates* the safety gate and *calibrates* $\hat{\delta}_{\text{UCB}}$; they are not reused below.

The finite- K mechanism, isolated. A controlled Gaussian illustration (design SD = transport SD, $\rho=0.9$; `fig_theory_illustration.py`) makes the source of the undercoverage explicit and shows why the deployed rule is a selector rather than a deconvolver. Both the naive and the finite- K -safe deconvolved score scales lie *below* the oracle s_R/s_{plug} at small K (Figure S1): subtracting a design-noise variance estimated from few countries over-shrinks the scale, and the resulting band is too narrow. The safe guard shrinks less, and both converge to the oracle as $K \rightarrow \infty$. The coverage consequence is severe and grows with the trajectory length L that the band must hold simultaneously (Figure S2): naive deconvolution falls to 0.43 at $K=25$, $L=16$, and the finite- K -safe scale improves it but does not by itself reach nominal at small K . The deployed adaptive selector detects that the remainder cannot be bounded and abstains to the conservative branch, remaining valid throughout at a width cost—the mechanism behind the $K=320$ -only activation in the confirmatory holdout in the main text.

S2 Extended robustness of the remainder bound

Cell-by-cell resolution of the base-method deficit. On the holdout the studentized band’s worst deployed cell was 0.843 (weak cross-round dependence at $K=25$), a finite- K self-inclusion deficit worst at small K and long trajectories. After switching to the unstudentized clustered band, a preregistered rerun on a fresh grid ($K \in \{15, \dots, 60\}$, ten DGPs) confirms the fix: the worst cell

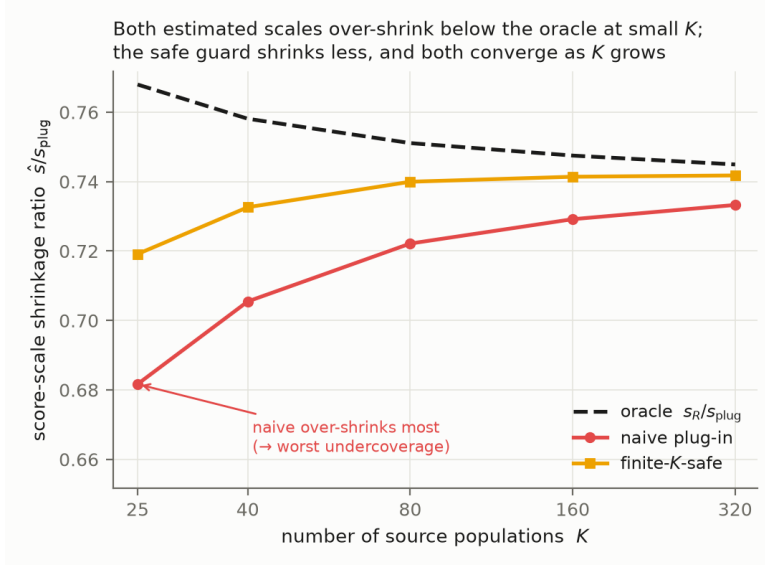


Figure S1: Score-scale shrinkage ratio $\hat{s}/s_{\text{plug}}$ vs K (Gaussian illustration, $\rho=0.9$, $L=8$). Naive and finite- K -safe deconvolution both over-shrink below the oracle at small K —the source of the finite- K undercoverage—with the safe guard shrinking less; both converge as K grows.

risks from 0.802 (S2, $K=15$) to 0.893, every cell meets its exact floor, and the width cost is 1.02–1.05 $\times$ on the real ESS transport bands (trust-CDF errors are $\approx$ homoskedastic across the low-trust core, so the constant-width band is competitive; the only cells where it pays more than 10% are synthetic heavy-tailed curves at $K=15$, where the studentized alternative is itself invalid).

Extreme-tail behavior of the remainder bound. Beyond the base-method deficit resolved by deploying the unstudentized clustered band, the holdout exposed a second, extreme-tail limitation. The Gaussian-calibrated remainder bound $\hat{\delta}_{\text{UCB}}$ is not robust at the extreme: in 77 of 360,000 draws (0.02%), on strongly-dependent or heavy-tailed DGPs at $K=320$ and $\rho \geq 0.72$, draws that pass gate B undercover to ≈ 0.79 ; because the gates route 96–98% of draws in those cells to conservative, the deployed cell coverage there is nonetheless 0.95–0.996.

S3 Additional empirical robustness

Design-preserving stress test. Subsampling the real AmericasBarometer while preserving its stratified-PSU structure, the estimated ρ saturates near 0.23: since $s_{\text{plug}}^2 = s_R^2 + \bar{v}^2$, shrinking the sample inflates design noise and observed spread in lockstep. The deconvolution regime is therefore unreachable even semi-synthetically at survey scale—direct evidence for the regime characterization of the main text.

The barriers are not survey artifacts. Both gates are distribution-free, so the boundary should transfer to any conformal procedure calibrated on estimated objects (the general remark of the main text). A simulation confirms it outside the survey setting (`e29_beyond_surveys`): across Gaussian, heavy-tailed, skewed, and a non-survey model-based small-area (MRP-style) calibration, the reliability floor $D \geq \sqrt{2/(K-1)}$ holds in every case (minimum $D/\text{floor} \geq 1.06$), and the gate opens only at $K \approx 130$ –150—if anything a stronger barrier than the floor-implied $K \geq 94$. The

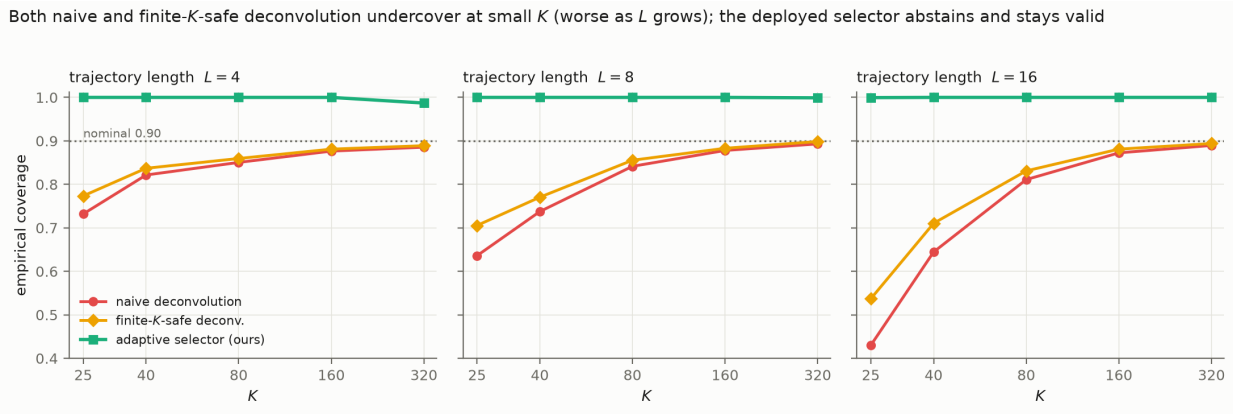


Figure S2: Latent-target coverage of the deconvolution branches vs K , faceted by trajectory length L (Gaussian illustration, $\rho=0.9$; nominal 0.90 dotted). Naive deconvolution undercovers severely at small K and worse as L grows; the finite- K -safe scale improves but does not itself reach nominal at small K . The deployed adaptive selector abstains to conservative and stays valid throughout.

one substantive difference is instructive: when a small area’s posterior noise is comparable to the between-area signal ($\hat{\rho}_{\text{LCB}} = 0.54$ at a noise-to-signal ratio of 0.5), the *need* gate does fire, unlike any cross-national survey. A many-area MRP application with appreciable posterior uncertainty is therefore the setting where the deconvolution could actually activate—the positive-result frontier the survey-scale impossibility leaves open.

The AmericasBarometer supplies the design effect the ESS lacks. The ESS releases design weights but limited cluster identifiers; the AmericasBarometer releases full stratified-PSU structure, letting us compute a proper design bootstrap. The design effect is real: the ratio of the proper stratified-PSU standard deviation to a naive independent-resampling one has median ≈ 1.13 – 1.20 and reaches 1.9 in the most clustered country-years. The proper band captures clustering variance the naive bootstrap misses. Yet certification is coarse-robust: the design effect moves band *width* but rarely flips the binary decline decision, because the decision is dominated by the between-country transport spread, not the within-country design noise—the same low- ρ fact, seen from the certification side.

Robustness across age groups (ESS). A preregistered age-group analysis (preregistered in docs/YOUTH_PREREGISTRATION.md) reruns the same certification within fixed age bands, changing nothing else. The “only Greece” result is not an artifact of pooling ages: it holds within the older (50+) group for both outcomes and within the middle (30–49) group for trust in parliament, so the persistent decline is concentrated in the older and middle adult population rather than created by averaging. Youth (18–29) certify *zero* persistent declines, but we read this as an underpowered null rather than youth stability: only 17 of 30 countries clear the preregistered minimum-sample gate for a single youth band, and the smaller, noisier youth cells yield wider design-aware bands that certify less—the small- K scope condition of the simulation appearing on real data. The analysis thus supports the headline while adding an honest caveat, and it is emphatically *not* a story of youth-led democratic disaffection.

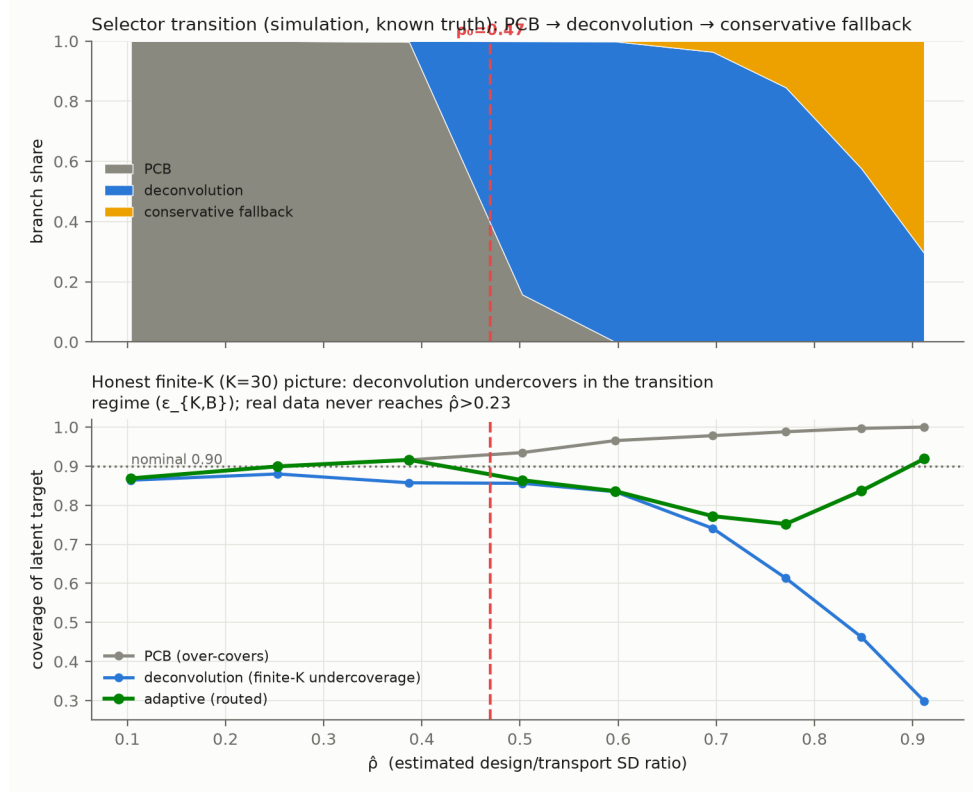


Figure S3: Development grid (Gaussian, known truth). The selector transitions $\text{PCB} \rightarrow \text{deconvolution} \rightarrow \text{conservative}$ as ρ grows; naive deconvolution undercovers at small K in the transition zone—the finite- K remainder the safety gate controls.

Youth deconsolidation on the World Values Survey. The youth claim in the Foa–Mounk reanalysis is only half-supported: the young show *more* persistent decline in rating democracy essential (7 vs. 3 countries) but *not* in openness to strongman or army rule (0–1 vs. 4), against the “youth embrace authoritarianism” reading.

References